URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf

RESOLUTION MSC.562(108)
(adopted on 23 May 2024)
STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR FISHING VESSEL PERSONNEL CODE (STCW-F CODE)

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECOGNIZING the importance of establishing detailed mandatory standards of competence and other mandatory provisions necessary to ensure that all fishing vessel personnel shall be properly educated and trained, adequately experienced, skilled and competent to perform their duties in a manner which provides for the safety of life and property at sea and the protection of the marine environment,

RECOGNIZING ALSO the need to allow for the timely amendment of such mandatory

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of the marine environment,
RECOGNIZING ALSO the need to allow for the timely amendment of such mandatory standards and provisions in order to effectively respond to changes in technology, operations, practices and procedures used on board fishing vessels,
BEARING IN MIND that a large percentage of maritime casualties and pollution incidents are caused by human error,
APPRECIATING that one effective means of reducing the risks associated with human error in the operation of seagoing fishing vessels is to ensure that the highest practicable standards of training, certification and competence are maintained in respect of fishing vessel personnel employed on such vessels,
DESIRING to achieve and maintain the highest practicable standards for the safety of life and property at sea and in port and for the protection of the environment,
NOTING resolution MSC.561(108), by which it adopted amendments to the annex to the
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NOTING resolution MSC.561(108), by which it adopted amendments to the annex to the International Convention on Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel, 1995 (1995 STCW -F Convention) which make the provisions of pa rt A of the Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel Code (STCW-F Code) mandatory under the Convention,
HAVING CONSIDERED, at its 108th session, the draft STCW -F Code, comprised of part A (Mandatory standards regarding provisions of the annex to the 1995 STCW-F Convention) and part B (Recommended guidance regarding provisions of the annex to the 1995 STCW -F Convention),
NOTING that regulation I/1.2 of the revised annex to the 1995 STCW -F Convention provides that part A of the STCW-F Code supplements the regulations annexed to the Convention and that any reference to a requirement in a regulation also constitutes a referen ce to the
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that any reference to a requirement in a regulation also constitutes a referen ce to the corresponding section of part A of the STCW-F Code,

1 ADOPTS:

.1 part A (Mandatory standards regarding provisions of the annex to the 1995 STCW-F Convention) of the Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel Code (STCW -F Code), set out in annex 1 to the present resolution;

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.2 part B (Recommended guidance regarding provisions of the annex to the 1995 STCW-F Convention) of the Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel Code (STCW -F Code), set out in annex 2 to the present resolution;

2 INVITES Parties to the 1995 STCW-F Convention to note:

.1 that the provisions of part A of the STCW -F Code will take effect for each

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2 INVITES Parties to the 1995 STCW-F Convention to note:

.1 that the provisions of part A of the STCW -F Code will take effect for each Party to the Convention on the same date and in the same manner as the amendments to the annex to that Convention adopted by resolution MSC.561(108);

.2 that the guidance contained in part B of the STCW-F Code should be taken into account by all Parties to the Convention as from the date of entry into force of the amendments to the annex to that Convention adopted by resolution MSC.561(108).

3 REQUESTS the Secretary -General to transmit certified copies of the present resolution and the text of the STCW -F Code contained in annexes 1 and 2 to all Parties to the 1995 STCW-F Convention;

4 ALSO REQUESTS the Secretary-General to transmit copies of this resolution and its annexes to Members of the Organization which are not Parties to the 1995 STCW -F Convention.

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annexes to Members of the Organization which are not Parties to the 1995 STCW -F Convention.

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STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR FISHING VESSEL PERSONNEL CODE (STCW-F CODE)

ANNEX 1

STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR FISHING VESSEL PERSONNEL CODE (STCW-F CODE)

Part A

Mandatory standards regarding provisions of the annex to the 1995 STCW-F Convention

Introduction

1 This part of the STCW -F Code contains mandatory provisions to which specific reference is made in the annex to the International Convention on Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel, 1995, as amended, hereinafter referred to as the 1995 STCW -F Convention. These requirements provide the minimum standards required to be maintained by Parties in order to give full and complete effect to the Convention.

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standards required to be maintained by Parties in order to give full and complete effect to the Convention.

2 Also contained in this part are standards of competence required to be demonstrated by candidates for the issue and revalidation of certificates of competency under the provisions of the 1995 STCW-F Convention. The abilities specified in the standards of competence are grouped, as appropriate, under the following seven functions:

.1-F Navigation
.2-F Catch handling and stowage
.3-F Controlling the operation of the vessel and care for persons on board
.4-F Marine engineering
.5-F Electrical, electronic and control engineering
.6-F Maintenance and repair
.7-F Radiocommunications

at the following levels of responsibility:

.1 Management level
.2 Operational level
.3 Support level

Functions and levels of responsibility are identified by the subtitle in the tables of standards of

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.2 Operational level
.3 Support level
Functions and levels of responsibility are identified by the subtitle in the tables of standards of competence given in chapter II of this part. The scope of the function at the level of responsibility stated in a subtitle is defined by the abilities listed under it in column 1 of the table. The meaning of "function" and "level of responsibility" is defined in general terms in section A-I/1 below.
3 The numbering of the sections of this part corresponds with the numbering of the regulations contained in the annex to the 1995 STCW-F Convention. The text of the sections may be divided into numbered parts and paragraphs, but such numbering is unique to that text alone.

Chapter I
Standards regarding general provisions
Section A-I/1
Definitions

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Chapter I
Standards regarding general provisions
Section A-I/1
Definitions
1 The definitions and clarifications contained in article II and regulation I/1 apply equally to the terms used in parts A and B of this Code. In addition, the following supplementary definitions apply only to this Code:
.1 Standard of competence means the level of proficiency to be achieved for the proper performance of functions on board vessels in accordance with the internationally agreed criteria as set forth herein and incorporating prescribed standards or levels of knowledge, understanding and demonstrated skill;
.2 Management level means the level of responsibility associated with:
.1 serving as skipper, chief engineer officer or second engineer officer on board a fishing vessel; and
.2 ensuring that all functions within the designated area of responsibility are properly performed;

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on board a fishing vessel; and
.2 ensuring that all functions within the designated area of
responsibility are properly performed;
.3 Operational level means the level of responsibility associated with:
.1 serving as officer in charge of a navigational or engineer watch or
as radio operator on board a fishing vessel; and
.2 maintaining direct control over the performance of all functions
within the designated area of responsibility in accordance with
proper procedures and under the direction of an individual serving
in the management level for that area of responsibility;
.4 Support level means the level of responsibility associated with performing
assigned tasks, duties or responsibilities on board a fishing vessel under the
direction of an individual serving in the operational or management level; and
.5 Evaluation criteria are the entries appearing in column 4 of the "Specification
of Minimum Standard of Competence" tables in part A and provide the means
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of Minimum Standard of Competence" tables in part A and provide the means
for an assessor to judge whether or not a candidate can perform the related
tasks, duties and responsibilities.
Section A-I/2
Application
(No provisions)
Section A-I/3
Certificates and endorsements
1 When provided in regulation I/3, paragraph 2, the certificate shall be issued in the
format 1 to 3, as shown below.
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STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR
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Format 1
The format used to attest the issue of a certificate shall be as shown below, provided that the
words "or until the date of expiry of any extension of the validity of this certificate as may be
shown overleaf" appearing on the front of the form and the provisions for recording extension
of the validity appearing on t he back of the form are omitted where the certificate is required
to be replaced upon its expiry.

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of the validity appearing on t he back of the form are omitted where the certificate is required to be replaced upon its expiry.

(Official seal)

(COUNTRY)

CERTIFICATE ISSUED UNDER THE PROVISIONS OF
THE INTERNATIONAL CONVENTION ON STANDARDS OF TRAINING,
CERTIFICATION AND WATCHKEEPING FOR FISHING VESSEL PERSONNEL, 1995

The Government of . . . . . certifies that the holder of this certificate has been found duly qualified in accordance with the provisions of regulation . . . . . of the above Convention and has been found competent to serve as specified below, subject to any limitations indicated until . . . . . or until the date of expiry of any extension of the validity of this certificate as may be shown overleaf.

The lawful holder of this certificate may serve in the following capacity or capacities:

CAPACITY LIMITATIONS APPLYING (IF ANY)

URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
The lawful holder of this certificate may serve in the following capacity or capacities:
CAPACITY LIMITATIONS APPLYING (IF ANY)
Certificate No. . . . . issued on . . . . . .
(Official seal)
. . . . . .
Signature of duly authorized official
. . . . . .
Name of duly authorized official
Date of birth of the holder of the certificate . . . . . .
Signature of the holder of the certificate . . . . . .
Photograph of the holder of the certificate
RESOLUTION MSC.562(108) (adopted on 23 May 2024)
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URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
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FISHING VESSEL PERSONNEL CODE (STCW-F CODE)
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The validity of this certificate is hereby extended until . . . . . .
(Official seal)
. . . . . .
Signature of duly authorized official
Date of revalidation . . . . . .
. . . . . .
Name of duly authorized official
The validity of this certificate is hereby extended until . . . . . .
(Official seal)
. . . . . .
Signature of duly authorized official
Date of revalidation . . . . . .
. . . . . .
Name of duly authorized official
RESOLUTION MSC.562(108) (adopted on 23 May 2024)
STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR

URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
Name of duly authorized official

Format 2
The form used to attest the issue of a certificate shall be as shown below, provided that the words "or until the date of expiry of any extension of the validity of this endorsement as may be shown overleaf" appearing on the front of the form and the provi sions for recording extension of the validity appearing on the back of the form are omitted where the endorsement is required to be replaced upon its expiry.
(Official seal)
(COUNTRY)
ENDORSEMENT ATTESTING THE ISSUE OF A CERTIFICATE
UNDER THE PROVISIONS OF THE INTERNATIONAL CONVENTION ON
STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING
FOR FISHING VESSEL PERSONNEL, 1995
The Government of . . . . . certifies that certificate no. . . . . .

URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf

The Government of . . . . . certifies that certificate no. . . . . .
has been issued to . . . . . who has been found duly qualified
in accordance with the provisions of regulation . . . . . of the above Convention
and has been found competent to serve as specified below, subject to any limitations indicated
until . . . . . or until the date of expiry of
any extension of the validity of this endorsement as may be shown overleaf.
The lawful holder of this endorsement may serve in the following capacity or capacities
specified in the applicable safe manning requirements of the Administration:
CAPACITY LIMITATIONS APPLYING (IF ANY)
Endorsement No. . . . . issued on . . . . . .
(Official seal)

URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf

Endorsement No. . . . . issued on . . . . . .

(Official seal)

. . . . . .

Signature of duly authorized official

. . . . . .

Name of duly authorized official

Date of birth of the holder of the certificate . . . . . .

Signature of the holder of the certificate . . . . . .

Photograph of the holder of the certificate

The validity of this endorsement is hereby extended until . . . . . .

(Official seal)

. . . . . .

URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf

(Official seal)

. . . . . .

Signature of duly authorized official

Date of revalidation . . . . . .

. . . . . .

Name of duly authorized official

The validity of this endorsement is hereby extended until . . . . . .

(Official seal)

. . . . . .

Signature of duly authorized official

Date of revalidation . . . . . .

. . . . . .

Name of duly authorized official

Format 3

The form used to attest the recognition of a certificate shall be as shown below, except that the words "or until the date of expiry of any extension of the validity of this endorsement as

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URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
the words "or until the date of expiry of any extension of the validity of this endorsement as may be shown overleaf" appearing on the front of the form and the p rovisions for recording extension of the validity appearing on the back of the form shall be omitted where the endorsement is required to be replaced upon its expiry.

(Official Seal)

(COUNTRY)

ENDORSEMENT ATTESTING THE RECOGNITION OF A CERTIFICATE UNDER THE PROVISIONS OF THE INTERNATIONAL CONVENTION ON STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR FISHING VESSEL PERSONNEL, 1995

The Government of . . . . . certifies that certificate no. . . . . issued to . . . . . by or on behalf of the Government of . . . . . is duly recognized in accordance with the provisions of regulation I/7 of the above Convention, and the lawful holder is authorized to serve as specified below, subject to any limitations

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and the lawful holder is authorized to serve as specified below, subject to any limitations indicated until . . . . . or until the date of expiry of any extension of the validity of this endorsement as may be shown overleaf.

The lawful holder of this endorsement may serve in the following capacity or capacities specified in the applicable safe manning requirements of the Administration:

CAPACITY LIMITATIONS APPLYING (IF ANY)

Endorsement No. . . . . issued on . . . . . .

(Official Seal)

. . . . . .

Signature of duly authorized official

. . . . . .

Name of duly authorized official

Date of birth of the holder of the certificate . . . . . .

URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
Date of birth of the holder of the certificate . . . . . .
Signature of the holder of the certificate . . . . . .
Photograph of the holder of the certificate

The validity of this endorsement is hereby extended until . . . . . .
(Official seal)
. . . . . .
Signature of duly authorized official
Date of revalidation . . . . . .
. . . . . .
Name of duly authorized official
The validity of this endorsement is hereby extended until . . . . . .
(Official seal)

URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
The validity of this endorsement is hereby extended until . . . . . .
(Official seal)
. . . . . .
Signature of duly authorized official
Date of revalidation . . . . . .
. . . . . .
Name of duly authorized official
RESOLUTION MSC.562(108) (adopted on 23 May 2024)
STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR
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Section A-I/4
Control procedures
(No provisions)
Section A-I/5
Communication of information
(No provisions)
Section A-I/6
Administration of certification arrangements
(No provisions)
Section A-I/7
Recognition of certificates
(No provisions)
Section A-I/8
Transitional provisions
(No provisions)
Section A-I/9
Dispensation
(No provisions)
Section A-I/10
Equivalents
(No provisions)
Section A-I/11
Use of simulators

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(No provisions)
Section A-I/9
Dispensation
(No provisions)
Section A-I/10
Equivalents
(No provisions)
Section A-I/11
Use of simulators
General performance standards for simulators used in training
1 Each Party shall ensure that any simulator used for simulator-based training shall:
.1 be suitable for the selected objectives and training tasks;
.2 be capable of simulating the operating capabilities of shipboard equipment concerned, to a level of physical realism appropriate to training objectives, and include the capabilities, limitations and possible errors of such equipment;
.3 have sufficient behavioural realism to allow a trainee to acquire the skills appropriate to the training objectives;
.4 provide a controlled operating environment capable of producing a variety of conditions, which may include emergency, hazardous or unusual situations relevant to the training objectives;
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relevant to the training objectives;

.5 provide an interface through which a trainee can interact with the equipment, the simulated environment and, as appropriate, the instructor; and
.6 permit an instructor to control, monitor and record exercises for the effective debriefing of the trainees.

General performance standards for simulators used in assessment of competence

2 Each Party shall ensure that any simulator used for the assessment of competence required under the Convention or for any demonstration of continued proficiency so required shall:
.1 be capable of satisfying the specified assessment objectives;
.2 be capable of simulating the operational capabilities of the shipboard equipment concerned to a level of physical realism appropriate to the
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.2 be capable of simulating the operational capabilities of the shipboard equipment concerned to a level of physical realism appropriate to the assessment objectives, and include the capabilities, limitations and possible errors of such equipment;

.3 have sufficient behavioural realism to allow a candidate to exhibit the skills appropriate to the assessment objectives;

.4 provide an interface through which a candidate can interact with the equipment and simulated environment;

.5 provide a controlled operating environment, capable of producing a variety of conditions, which may include emergency, hazardous or unusual situations relevant to assessment objectives; and

.6 permit an assessor to control, monitor and record exercises for the effective assessment of the performance of candidates.

Additional performance standards

3 In addition to meeting the basic requirements set out in paragraphs 1 and 2, simulation

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Additional performance standards

3 In addition to meeting the basic requirements set out in paragraphs 1 and 2, simulation equipment to which this section applies shall meet the performance standards given below in accordance with their specific type.

Radar simulation

4 Radar simulation equipment shall be capable of simulating the operational capabilities of navigational radar equipment which meets all applicable performance standards adopted by the Organization and incorporate facilities to:

.1 operate in the stabilized relative -motion mode and sea - and ground-stabilized true-motion modes;

.2 model weather, tidal streams, current, shadow sectors, spurious echoes and other propagation effects, and generate coastlines, navigational buoys and search and rescue transponders; and

.3 create a real -time operating environment incorporating at least two own-vessel stations with ability to change the own vessel's course and

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.3 create a real -time operating environment incorporating at least two own-vessel stations with ability to change the own vessel's course and speed, and include parameters for at least 20 target vessels and appropriate communication facilities.

Simulator training objectives

5 Each Party shall ensure that the aims and objectives of simulator -based training are defined within an overall training programme and that specific training objectives and tasks are selected so as to relate as closely as possible to shipboard tasks and practices.

Training procedures

6 In conducting simulator-based training, instructors shall ensure that:

.1 trainees are adequately briefed beforehand on the exercise objectives and tasks and are given sufficient planning time before the exercise starts;

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tasks and are given sufficient planning time before the exercise starts;
.2 trainees have adequate familiarization time on the simulator and with its equipment before any training or assessment exercise commences;
.3 guidance given and exercise stimuli are appropriate to the selected exercise objectives and tasks and to the level of trainee experience;
.4 exercises are effectively monitored, supported as appropriate by audio and visual observation of trainee activity and pre - and post-exercise evaluation reports;
.5 trainees are effectively debriefed to ensure that training objectives have been met and that operational skills demonstrated are of an acceptable standard;
.6 the use of peer assessment during debriefing is encouraged; and
.7 simulator exercises are designed and tested so as to ensure their suitability for the specified training objectives.

Assessment procedures

7 Where simulators are used to assess the ability of candidates to demonstrate levels

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for the specified training objectives.

Assessment procedures

7 Where simulators are used to assess the ability of candidates to demonstrate levels of competency, assessors shall ensure that:

.1 performance criteria are identified clearly and explicitly and are valid and available to the candidates;

.2 assessment criteria are established clearly and are explicit to ensure reliability and uniformity of assessment and to optimize objective measurements and evaluation, so that subjective judgements are kept to the minimum;

.3 candidates are briefed clearly on the tasks and/or skills to be assessed and on the tasks and performance criteria by which their competency will be determined;

.4 assessment of performance takes into account normal operating procedures and any behavioural interaction with other candidates on the simulator or with simulator staff;

.5 scoring or grading methods to assess performance are used with caution until they have been validated; and

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with simulator staff;
.5 scoring or grading methods to assess performance are used with caution until they have been validated; and
.6 the prime criterion is that a candidate demonstrates the ability to carry out a task safely and effectively to the satisfaction of the assessor.

Section A-I/12
Medical standards
1 Parties, when establishing standards of medical fitness for fishing vessel personnel as required by regulation I/12, shall adhere to the minimum in -service eyesight standards set out in table A-I/12 and take into account the criteria for physical and medical fitness set out in paragraph 2. They should also take into account the guidance given in section B -I/12 of this Code and the Guidelines on the medical examination of fishing vessel personnel . These
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Code and the Guidelines on the medical examination of fishing vessel personnel . These standards may, to the extent determined by the Party without prejudice to the safety of the fishing vessel personnel or the fishing vessel, differentiate between those persons seeking to start a career at sea and those fishing vessel personnel already serving at sea and between different functions on board, bearing in mind the different duties of fishing vessel personnel. They shall also take into account any impairment or disease that will limit the ability of the fishing vessel personnel to effectively perform their duties during the validity period of the medical certificate.

2 The standards of physical and medical fitness established by the Party shall ensure that fishing vessel personnel satisfy the following criteria:

.1 have the physical capability, taking into account paragraph 5 below to fulfil all the requirements of the basic safety training as required by section A-III/1;

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all the requirements of the basic safety training as required by section A-III/1;
.2 demonstrate adequate hearing and speech to communicate effectively and detect any audible alarms;
.3 have no medical condition, disorder or impairment that will prevent the effective and safe conduct of their routine and emergency duties on board during the validity period of the medical certificate;
.4 are not suffering from any medical condition likely to be aggravated by service at sea or to render the fishing vessel personnel unfit for such service or to endanger the health and safety of other persons on board; and
.5 are not taking any medication that has side effects that will impair judgement, balance or any other requirements for effective and safe performance of routine and emergency duties on board.

3 Medical fitness examinations of fishing vessel personnel shall be conducted by appropriately qualified and experienced medical practitioners recognized by the Party.

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appropriately qualified and experienced medical practitioners recognized by the Party.
4 Each Party shall establish provisions for recognizing medical practitioners. A register of recognized medical practitioners shall be maintained by the Party and made available to other Parties, companies and fishing vessel personnel on request.
5 Each Party shall provide guidance for the conduct of medical fitness examinations and issuing of medical certificates, taking into account the guidance given in section B-I/12 of this Code and the Guidelines on the medical examination of fishing vessel personnel . Each Party shall determine the amount of discretion given to recognized medical practitioners on the application of the medical standards, bearing in mind the different duties of fishing vessel personnel, except that there shall not be discretion with respect to the minimum eyesight standards for distance vision aided, near/immediate vision and colour vision in table A-I/12 for
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standards for distance vision aided, near/immediate vision and colour vision in table A-I/12 for fishing vessel personnel in the deck department required to undertake lookout duties. A Party may allow discretion on the application of these standards with regard to fishing vessel personnel in the engine department, on the condition that fishing vessel personnel's combined vision fulfils the requirements set out in table A-I/12.

RESOLUTION MSC.562(108) (adopted on 23 May 2024)

STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR FISHING VESSEL PERSONNEL CODE (STCW-F CODE)

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RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF
LIQUEFIED HYDROGEN IN BULK

ANNEX 20
RESOLUTION MSC.565(108)
(adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF
LIQUEFIED HYDROGEN IN BULK
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
NOTING that the International Convention for the Safety of Life at Sea ("the Convention"), 1974 and the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk ("the IGC Code") currently do not specifically provide requirements for carriage of liquefied hydrogen in bulk by sea,
NOTING ALSO that paragraph 5 of Preamble of the IGC Code states that requirements for new products and their conditions of carriage will be circulated as recommendations, on an
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new products and their conditions of carriage will be circulated as recommendations, on an interim basis, prior to the entry into force of the appropriate amendments,
RECOGNIZING a need for the development of the Revised interim recommendations for carriage of liquefied hydrogen in bulk,
ACKNOWLEDGING that, in the interim, there is an urgent need to provide recommendations to the Administrations on safe carriage of liquefied hydrogen in bulk,
ACKNOWLEDGING ALSO that the Revised Interim Recommendations are intended to facilitate establishment of a tripartite agreement for a pilot ship, which will be developed for the research and demonstration of safe long-distance overseas carriage of liquefied hydrogen in bulk,
HAVING CONSIDERED the Revised Interim Recommendations prepared by the Sub - Committee on Carriage of Cargoes and Containers at its ninth session,
1 ADOPTS the Revised interim recommendations for carriage of liquefied hydrogen in
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1 ADOPTS the Revised interim recommendations for carriage of liquefied hydrogen in bulk, the text of which is set out in the annex to the present resolution;
2 INVITES Member States to apply the Revised Interim Recommendations to the pilot ship carrying liquefied hydrogen in bulk taking the explanatory notes into consideration;
3 AGREES to acquire information on safe carriage of liquefied hydrogen in bulk prior to amendment to the IGC Code for the inclusion of liquefied hydrogen;
4 ALSO AGREES that these Revised Interim Recommendations may need to be reviewed if they are to be applied to ships other than the pilot ship; and
5 URGES Member States and the industry to submit information, observations, comments and recommendations based on the practical experience gained through the application of the Revised Interim Recommendations and submit relevant safety analysis on ships carrying liquefied hydrogen in bulk.
6 REVOKES resolution MSC.420(97).

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on ships carrying liquefied hydrogen in bulk.
6 REVOKES resolution MSC.420(97).
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ANNEX
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE
OF LIQUEFIED HYDROGEN IN BULK
1 INTRODUCTION
1.1 For the carriage of liquefied gases in bulk by ships, the ships should comply with the relevant requirements in the IGC Code, as amended ("the Code"). The scope of the Code provided in paragraph 1.1.1 is:
"The Code applies to ships regardless of their size, including those of less than 500 gross tonnage, engaged in the carriage of liquefied gases having a vapour pressure exceeding 0.28 MPa absolute at a temperature of 37.8°C, and other products, as shown in chapter 19, when carried in bulk".
1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen
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products, as shown in chapter 19, when carried in bulk".
1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen carrier") should comply with the Code.
1.3 The Code requires that a gas carrier should comply with the minimum requirements for the cargo listed in chapter 19. However, the requirements for liquefied hydrogen are not specified in the Code.
1.4 This annex provides the Revised Interim Recommendations, as referred to in paragraph 5 of the preamble of the Code, for the carriage of liquefied hydrogen in bulk, which are intended to provide the basis for the future minimum requirements for the carriage of this cargo. The Revised Interim Recommendations are intended to facilitate the establishme nt of a tripartite agreement among the relevant Administrations for the carriage of liquefied hydrogen in bulk. However, they are not intended to prohibit the adoption of designs and arrangements
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in bulk. However, they are not intended to prohibit the adoption of designs and arrangements other than those specified in the Code or in these Recommendations, at the discretion of the Administrations.

1.5 These recommendations have been developed under the assumption that a liquefied hydrogen carrier does not carry liquefied gases other than liquefied hydrogen. These recommendations, therefore, are not applicable to liquefied hydrogen carriers carrying gase s other than liquefied hydrogen.

1.6 In the Code, reference is made to paragraph 5 of the Preamble; paragraph 1.1.6.1; and Note No.8 on completion of certificate in "model form of international certificate of fitness for the carriage of liquefied gases in bulk" in appendix 2 to the Code.

1.7 These Revised Interim Recommendations consist of the following parts. Part A is applicable to ships with any type of cargo containment system. Part B and subsequent part(s)

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applicable to ships with any type of cargo containment system. Part B and subsequent part(s) prescribe additional special requirements for cargo containment systems of specific types.

Part A: General (applicable to ships with any type of cargo containment system);

Part B: Cargo containment systems of independent cargo tanks using vacuum insulation; and

Part C: Cargo containment systems of independent cargo tanks using insulation materials and hydrogen gas in the inner insulation spaces.

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1.8 Part A of this document was developed based on the design of parts B and C. If subsequent part(s) are added, the special requirements prescribed in part A may be reviewed.

Part A

General

(Applicable to ships with any type of cargo containment system)

2 GENERAL

2.1 Definition

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reviewed.
Part A
General
(Applicable to ships with any type of cargo containment system)
2 GENERAL
2.1 Definition
2.1.1 The following definition should apply for the purpose of these Revised Interim Recommendations.
Permeation is flow of a fluid through another material by diffusion without a defect or opening of the latter.8
2.2 Requirements for carriage of liquefied hydrogen in bulk
2.2.1 The requirements for the carriage of liquefied hydrogen in bulk have been developed based on the results of a comparison study of similar cargoes listed in chapter 19 of the Code, e.g. liquefied natural gas.
2.2.2 Chapter 19 of the Code governs the application of general requirements for respective cargoes. Selections of the general requirements for respective cargoes are expressed in columns 'c' to 'g'. In addition to general requirements, special requirement s may apply to specific cargoes depending on the properties/hazards of the cargoes.

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specific cargoes depending on the properties/hazards of the cargoes.
2.2.3 Tables 1 and 2 specify the proposed selection of the general requirements and the special requirements, respectively, for liquefied hydrogen. In addition to table 2, special requirements for cargo containment systems of specific types are prescribed in part B or subsequent part(s).
8 1 See paragraph 3.79 of ISO/TR 15916:2015.
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Table 1: Requirements for carriage of liquefied hydrogen in bulk

a	b	c	d	e	f	g	h	i
Product name	Ship type	Independent tank type C required	Control of vapour space within cargo tanks	Vapour detection	Gauging	Special requirements		
Hydrogen	2G	-	-	F	C	See table 2 and, either corresponding table 4 or table 5, as appropriate for the type of cargo containment systems		

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Hydrogen 2G - - F C
See table 2
and, either
corresponding
table 4 or
table 5, as
appropriate for
the type of
cargo
containment
systems

Table 2: Special requirements for carriage of liquefied hydrogen in bulk

No.	Special requirement	Related hazard
A-1	Requirements for materials whose design temperature is lower than -165°C should be agreed with the Administration, paying attention to appropriate standards. Where minimum design temperature is lower than -196°C, property testing for insulation materials should be carried out with the appropriate medium, over a range of temperatures expected in service.	Low temperature (see 4.2.1)
A-2	Materials of construction and ancillary equipment such as insulation should be resistant to the effects of high oxygen concentrations caused by condensation and enrichment at the low temperatures attained in parts of the cargo system (refer to the requirement for nitrogen). This special requirement is applied	---

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low temperatures attained in parts of the cargo system (refer to the requirement for nitrogen). This special requirement is applied to all locations where contact with condensed oxygen is anticipated under normal conditions and foreseeable single failure scenarios.

Low temperature
(see 4.2.2)

A-3 For cargo pipes containing liquid hydrogen and cold hydrogen vapour, measures should be taken to prevent the exposed surfaces from reaching -183°C. For places where preventive measures against low temperature are not sufficiently effective, such as cargo m anifolds, other appropriate measures such as ventilation which avoids the formation of highly enriched oxygen and the installation of trays recovering liquid air may be permitted in lieu of the preventive measures. Insulation on liquid hydrogen piping syst ems exposing to air should be of non -combustible material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent

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material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent oxygen enrichment within the insulation.
Low temperature
(see 4.2.2)
A-4 Appropriate means, e.g. filtering, should be provided in cargo piping systems to remove impure substances condensed at low temperature.
Low temperature
(see 4.2.3)
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No. Special requirement Related hazard
A-5 Pressure relief systems should be suitably designed and constructed to prevent blockage due to formation of water or ice.
Low temperature
(see 4.2.4)
A-6 At places where contact with hydrogen is anticipated, suitable materials should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and
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materials should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and fatigue properties due to continual exposure to hydrogen , as necessary.

Hydrogen embrittlement (see 4.3)

A-7 Double tube structures ensuring no leakage, or fixed hydrogen detectors being capable of detecting a hydrogen leak, should be provided for confined places where leakage of hydrogen may occur, such as cargo valves, flanges and seals.

Susceptibility to leakage (see 4.4.2)

A-8 Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the tightness test medium for cargo tank and cargo piping.

Susceptibility to leakage (see 4.4.3)

A-9 The amount of carbon dioxide carried for a carbon dioxide fire-extinguishing system should be sufficient to provide a quantity of free gas equal to 75% or more of the gross volume of the cargo compressor and pump rooms in all cases.

Fire by Hydrogen (see 4.7.3)

Wide range of

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the cargo compressor and pump rooms in all cases.
Fire by Hydrogen
(see 4.7.3)
Wide range of
flammability limits
(see 4.10)
A-10 When deterioration of insulation capability by single damage is possible, appropriate safety measures should be adopted taking into account the deterioration.
High pressure
(see 4.8)
A-11 Appropriate measures should be provided to prevent vents becoming blocked by accumulations of ice formed from moisture in the air.
Low temperature
(see 4.2.2)
A-12 Due consideration should be given to means for handling boil-off gas.
High pressure
(see 4.8)
A-13 Due consideration should be given to static electricity associated with rotating or reciprocating machinery including the installation of conductive machinery belts and precautionary measures incorporated in operating and maintenance procedures , in addition to the bonding of tanks, piping and equipment required by paragraph 5.7.4 of the Code.
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addition to the bonding of tanks, piping and equipment required by paragraph 5.7.4 of the Code.
Anti-static clothing and footwear, and a portable hydrogen detector should be provided for each crew member working in the cargo area.
Static electricity
(see 4.9.2)
A-14 The cargo operation manuals required in paragraph 18.2 of the Code should include limitations of various operations in relation to environmental conditions.
Wide range of
flammability limits
(see 4.10)
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No. Special requirement Related hazard
A-15 An appropriate procedure should be established for warm -up, inert gas purge, gas -free, hydrogen purge and pre -cooling.
The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;

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The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;
.3 measurement of temperature;
.4 rates of supply of gases;
.5 conditions for commencement, suspension, resuming and termination of each operation;
.6 treatment of return gases; and
.7 discharge of gases.

Prevention of dangerous purging operation (see 4.11)

A-16 Only almost pure para-hydrogen (i.e. more than 95%) should be loaded in order to avoid excessive heating by ortho - to para-hydrogen conversion.

General (see 4.1)

A-17 Fire detectors for detecting hydrogen fire should be selected, taking into account the features of hydrogen fire, to the satisfaction of the Administration.

Features of hydrogen fire and fire hazard (see 4.7.4)

A-18 At the design stage, dispersion of hydrogen from vent outlets should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces,

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should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces, machinery spaces and control stations. Extension of hazardous areas should be considered based on the results of the analysis.

Low density and high diffusivity (see 4.5)

A-19 Due consideration should be given to appropriate safety measures to prevent formation of explosive mixture in the case of a leakage and permeation of hydrogen, including:

.1 installation of hydrogen detectors in order to detect a possible ground -level travel of low temperature hydrogen gas, and at high points in spaces where warm hydrogen gas can be trapped; and

.2 application of "best practice" for land -based liquid hydrogen storage taking into account appropriate guidance such as "Cryogenics Safety Manual Fourth Edition (1998)"8).

General (see 4.1)

A-20 In the case that fusible elements are used as a means of fire

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Edition (1998)"8).
General
(see 4.1)
A-20 In the case that fusible elements are used as a means of fire detection required by paragraph 18.10.3.2 of the Code, flame detectors suitable for hydrogen flames should be provided in addition at the same locations. Appropriate means should be adopted to prevent the activation of ESD system owing to false alarm of flame detectors, e.g. avoiding activation of ESD system by single sensor (voting method).
Fire hazard
(see 4.7.4)
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No. Special requirement Related hazard
A-21 Consideration should be given to enhance the ventilation capacity of the enclosed spaces subject to liquefied hydrogen leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in
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leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in relation to temperature and heat capacity of adjacent spaces.

Low density and high diffusivity (see 4.5)

A-22 Liquid and gas hydrogen pipes should not pass through enclosed spaces in addition to other than those referred to in paragraph 5.2.2.1.2 of the Code, unless:

.1.1 the spaces are equipped with gas detection systems which activate the alarm at not more than 20% LFL and shut down the isolation valves, as appropriate, at not more than 40% LFL (see sections 16.4.2 and 16.4.8 of the Code); and

.1.2 the spaces are adequately ventilated; or

.2 the spaces are maintained in an inert condition.

Susceptibility to leakage (see 4.4)

A-23 A risk assessment should be conducted to ensure that risks arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship

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arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship are addressed. Consideration should be given to the hazards associated with properties of liquefied hydrogen and hydrogen gas, physical layout, operation and maintenance, following any reasonably foreseeable failure. For the risk assessment, appropriate methods, e.g. HAZID, HAZOP, FMEA/FMECA, what-if analysis, etc., should be adopted taking into account IEC/ISO 31010:2019 "Risk management Risk assessment techniques"7) and SAE ARP 5580 -2001 "Recommended failure modes and effects analysis (FMEA) practices for non-automobile applications"9).

General

(see 4.1)

A-24 Relief valve sizing should be undertaken for the most onerous scenario. The evaluation should include the fire scenario and should consider the resulting magnitude of the heat flux on the cargo containment system.

High pressure

hazard

(see 4.8)

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should consider the resulting magnitude of the heat flux on
the cargo containment system.
High pressure
hazard
(see 4.8)
A-25 A filling limit exceeding 98% at reference temperature should not
be permitted.
High pressure
hazard
(see 4.8)
A-26 Bolted flange connections of hydrogen piping should be avoided
where welded connections are feasible.
Susceptibility to
leakage (see
4.4.2)
A-27 Due consideration should be given to the invisible nature of
hydrogen fire from the viewpoint of safety of ships and especially
personnel in case of fire.
Fire hazard
(see 4.7.1)
3 EXPLANATION ON GENERAL REQUIREMENTS
3.1 Properties of liquefied hydrogen
The application of general requirements in the Code for liquefied hydrogen has been considered based on a comparison study on the physical properties of liquefied hydrogen and LNG. LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate
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and LNG. LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate
flammable high pressure gas. For reference, table 3 shows the comparison of physical
properties of hydrogen and methane, the major component of LNG.
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Table 3: Comparison of physical properties of Hydrogen and Methane
Hydrogen Methane References
Boiling temperature (K)* 20.3 111.6 ISO1), Annex A, Table A.3
Liquid density (kg/m3)* 70.8 422.5 ISO1), Annex A, Table A.3
Gas density (kg/m3)** (Air: 1.198) 0.084 0.668 NIST RefProp10)
Viscosity (g/cms x 10-6)
Gas
Liquid
8.8
13.49
10.91
116.79
NIST RefProp10)
NIST RefProp10)
Flame temperature in air (°C) 2396 2230 Calculated using Cantera and GRI 3.0 mechanism
Maximum burning velocity (m/s) 3.15 0.385 Calculated using Cantera and GRI 3.0 mechanism
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Maximum burning velocity (m/s) 3.15 0.385 Calculated using Cantera and GRI 3.0 mechanism
Heat of vaporization (J/g)* 454.6 510.4 ISO1), Annex A, Table A.3
Lower flammability limit
(% vol. fraction)*** 4 5.3 ISO1), Annex B, Table B.2
Upper flammability limit
(% vol. fraction)*** 77 17.0 ISO1), Annex B, Table B.2
Minimum ignition energy (mJ)*** 0.017 0.274 ISO1), Annex B, Table B.2
Auto-ignition temperature*** 858 810 ISO1), Annex B, Table B.2
Toxicity Non Non Orange book5)
Temperature at critical point (K) 33.19**** 190.55
Hydrogen: ISO1), Annex A,
Table A.1
Methane: The Japan
Society of Mechanical
Engineers, Data Book,
Thermophysical
Properties of Fluids (1983)
Pressure at critical point (kPaA) 1315**** 4595
Hydrogen: ISO1), Annex A,
Table A.1
Methane: The Japan
Society of Mechanical
Engineers, Data Book,
Thermophysical
Properties of Fluids (1983)
Remarks: * At their normal boiling points for comparison purpose.
** At normal temperature and pressure.

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Properties of Fluids (1983)
Remarks: * At their normal boiling points for comparison purpose.
** At normal temperature and pressure.
*** Ignition and combustion properties for air mixtures at 25°C
and 101.3 kPaA.
**** Normal Hydrogen.
3.2 Explanation on respective requirements
3.2.1 Ship type (column 'c')
As the hazard associated with hydrogen cargo is flammability but not toxicity, the ship type is
considered 2G.
3.2.2 Independent tank type C required (column 'd')
Independent tank type C is allocated only to dangerous goods of class 2.3 whose vapour density
is heavier than air. Independent tank type C is considered not to be required for liquefied hydrogen.
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3.2.3 Control of vapour space within cargo tank (column 'e')

3.2.3 Control of vapour space within cargo tank (column 'e')

Special environment controls such as drying and inerting are generally required for liquid chemical products in consideration of the reactivity of cargo vapour and air. As is the case for LNG, it is considered not to be necessary to apply such requirements for liquefied hydrogen.

3.2.4 Vapour detection (column 'f')

Because hydrogen is flammable and non -toxic, it is appropriate to require Flammable (F) as vapour detection for liquefied hydrogen.

3.2.5 Gauging (column 'g')

On the grounds that Closed (C) gauging is required, in principle, for flammable or toxic cargoes, such as methane, it is considered to be appropriate to require Closed (C) gauging for hydrogen, taking into account that hydrogen has high ignitability and a wide flammable range in air and that closed gauging is effective to prevent leakage of gases into air.

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that closed gauging is effective to prevent leakage of gases into air.

4 SPECIAL REQUIREMENTS AGAINST HAZARDS OF LIQUEFIED HYDROGEN

4.1 Hazards of liquefied hydrogen to be considered

4.1.1 The hazards related to liquefied hydrogen are low ignition energy , a wide range of flammability limits, low visibility of flames in case of fire, high flame velocity which may lead to the detonation with shockwave, low temperature and liquefaction/solidification of inert gas and constituents of air which may result in an oxygen-enriched atmosphere, high permeation, low viscosity, and hydrogen embrittlement including weld metals. Where vacuum insulation is adopted, due consideration should be given to the possibility of untimely deterioration of insulation properties at the envisaged carriage temperatures of liquid hydrogen. The vacuum insulation evaluation should be specified for the normal range or upper limit of cold vacuum

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insulation evaluation should be specified for the normal range or upper limit of cold vacuum pressure (CVP), and loss of vacuum should be defined with respect to this value. Accordingly, effect of vacuum pressure should be taken into account at the time of design and testing of piping with vacuum insulation. Supporting structure and adjacent hull structure should be designed taking into account the cooling owing to loss of vacuum insulation.

4.1.2 Hydrogen is essentially a mixture of ortho - and para-hydrogen, with an equilibrium concentration of 75% ortho-hydrogen and 25% para-hydrogen at ambient temperature. When liquefied at 20K, there is a slow but continuous transformation of ortho -hydrogen to para-hydrogen. The exothermic conversion of the nuclear spin isomers of hydrogen (ortho- to para-hydrogen) may take place and the effect of the conversion may have an impact on the cooling capacity and relief valve capacity of the vessel's equipment.

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the cooling capacity and relief valve capacity of the vessel's equipment.
4.1.3 Trace amounts of air will condense or solidify in an environment with liquid hydrogen possibly resulting in an unstable and explosive mixture. Precautions should be taken to assure that the possibility of condensed air is accounted within properly secured hazard areas.
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4.2 Low temperature hazard
4.2.1 Selection of appropriate material
4.2.1.1 Tables 6.3 and 6.4 in the Code prescribe material selection for piping or cargo tanks whose design temperature is -165°C or higher. According to Note 2 of table 6.3 and Note 3 of table 6.4 of the Code, the requirements for materials whose design temperatures are lower than -165°C should be specially agreed with the Administration. In this regard, the publication
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than -165°C should be specially agreed with the Administration. In this regard, the publication by AIAA2) introduces some appropriate materials corresponding to the design temperature and the Administration should take into account such references for the material selection.

4.2.1.2 Although paragraph 4.19.3 in the Code requires testing of materials used for thermal insulation for various properties adequate for the intended service temperature, the minimum test temperature is -196°C. The requirements in the Code do not refer to the normal boiling point of hydrogen, being -253°C. In case of carriage of liquefied hydrogen, special requirements should be provided to consider the lower design temperature.

4.2.2 Measures for condensed air

4.2.2.1 In the case of nitrogen whose normal boiling point is -196°C, for which air condensation and oxygen enrichment are concerns, the following special requirement has already been included in paragraph 17.17 in the Code:

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condensation and oxygen enrichment are concerns, the following special requirement has already been included in paragraph 17.17 in the Code:

"Material of construction and ancillary equipment such as insulation shall be resistant to the effect of high oxygen concentrations caused by condensation and enrichment at the low temperatures attained in parts of the cargo system. Due consideration shall be given to ventilation in such areas where condensation might occur to avoid the stratification of oxygen-enriched atmosphere."

A similar special requirement is applicable to hydrogen.

4.2.2.2 A vent may be blocked by accumulation of ice formed from moisture in the air, which may result in excessive pressure leading to rupture of the vent and relevant piping (see paragraph 4.2.4).

4.2.3 Removal of impure substances condensed

The removal of impure substances, such as those contained in condensate in pipes, should

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4.2.3 Removal of impure substances condensed

The removal of impure substances, such as those contained in condensate in pipes, should be separately considered. Installation of filters can be an appropriate measure and should be stipulated as a special requirement.

4.2.4 Prevention of blockage due to formation of water or ice

Pressure relief systems may become blocked due to formation of water or ice, depending on the temperature and humidity of air, resulting from the low temperature of the cargo and its vapour (see paragraph 4.2.2) . Appropriate means should be provided to prevent such phenomena.

4.3 Hydrogen embrittlement

4.3.1 Selection of appropriate materials should be required to prevent failures owing to hydrogen embrittlement. The publication by AIAA 2) introduces some appropriate materials resistant to hydrogen embrittlement, and concludes that aluminium is the material least affected.

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resistant to hydrogen embrittlement, and concludes that aluminium is the material least affected.

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4.3.2 International or national standards should be followed for the selection of materials for the design of liquefied and gaseous hydrogen installations in a marine environment.

4.4 Susceptibility to leakage

4.4.1 Prevention of leakage from pipes

To mitigate undetected accumulation of hydrogen in a confined space, effective measures should be employed to reduce the possibility of leakage of hydrogen, taking its leakage characteristics into account. Effective measures can be double tube structures, or fixed hydrogen leak detectors in areas assessed as being highly hazardous with regard to hydrogen

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hydrogen leak detectors in areas assessed as being highly hazardous with regard to hydrogen leakage. Hydrogen leakage through welds, joints and seals is an important consideration for the design of hydrogen systems and an important operational issue.

4.4.2 Implementation of effective tightness test

4.4.2.1 Tightness tests for cargo tanks and cargo pipes/valves are required by paragraphs 4.20.3.2, 5.13.1 and 5.13.2.3 in the Code respectively. Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the medium for tightness tests, instead of air, because hydrogen is highly susceptible to leakage.

4.4.2.2 For a hydrogen installation, the pipework should be pressure -tested at its design pressure. Consideration should be given to using oxygen-free nitrogen with a small molecule tracer gas, such as helium as the test medium and an electronic leak detec tor for identifying leaks.

4.4.3 Confirmation of appropriate operating procedure

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leaks.

4.4.3 Confirmation of appropriate operating procedure

Instructions/manuals containing the operating procedures for the prevention of leakage during transport, methods for early detection in case of leakage, and appropriate measures after such events, should be provided. For this, paragraph 18.3 of the Code requires that the information shall be on board and available to all concerned, giving the necessary data for the safe carriage of cargo. In detail, the Code requires such information on action to be taken in the event of spills or leak, countermeasures agai nst accidental personal contact, procedures for cargo transfer, and emergency procedures to be on board. With regard to the manuals on procedures for liquefied hydrogen during carriage and transfer operations, the requirements in the Code are applicable and no special requirement is necessary.

4.5 Low density and high diffusivity

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are applicable and no special requirement is necessary.

4.5 Low density and high diffusivity

Though low density and high diffusivity of hydrogen may reduce the possibility of formation of a flammable atmosphere in open spaces, adequate ventilation is necessary for enclosed spaces in cargo areas where formation of hydrogen -oxygen/air mixture may occur. Paragraph 12.2 of the Code requires fixed ventilation systems or portable mechanical ventilation for such enclosed spaces. These requirements in the Code are applicable to liquefied hydrogen carriers and no special requirement is necessary in this regard.

4.6 Ignitability

4.6.1 The Code requires electrical bonds of the piping and the cargo tanks in paragraph 5.7.4, exclusion of all sources of ignition in paragraph 11.1.2, electrical installations to minimize the risk of fire and explosion from flammable products in paragraph 10.2.1 and so on, in order to prevent ignition of flammable cargoes.

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on, in order to prevent ignition of flammable cargoes.
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4.6.2 The Code requires compliance with the relevant standards issued by the International Electrotechnical Commission (IEC) and the IEC standards specify the details of such safety measures depending on the respective properties of flammable gases including hydrogen. No special requirement is necessary with regard to ignitability of hydrogen.*
4.7 Fire hazard
4.7.1 Safety of personnel in case of fire
To avoid the effects of flame and UV radiation produced by a hydrogen fire, it is effective to use fire-fighter's outfits and protective equipment. The Code already requires fire -fighter's outfits for ships carrying flammable products in paragraph 11.6.1 and safety equipment in
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outfits for ships carrying flammable products in paragraph 11.6.1 and safety equipment in paragraph 14.3. This issue should be considered as the matter of cargo information required by paragraph 18.3 of the Code. Due consideration should be given to the invisible nature of hydrogen fire.

4.7.2 Compatibility of fire-extinguishing systems

Dry chemical powder fire -extinguishing or carbon dioxide fire -extinguishing systems are considered to be effective in case of hydrogen fire and such fire -extinguishing systems are already required by paragraphs 11.4 and 11.5 of the Code. Special requiremen ts for installation of other types of fire -extinguishing systems are considered unnecessary, except with regard to the increased of amount of carbon dioxide required, as mentioned in the next paragraph in this document.

4.7.3 Increase of the amount of gas for carbon dioxide fire-extinguishing systems

4.7.3.1 Paragraph 11.5.1 of the Code requires as follows:

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4.7.3 Increase of the amount of gas for carbon dioxide fire-extinguishing systems

4.7.3.1 Paragraph 11.5.1 of the Code requires as follows:

"Enclosed spaces meeting the criteria of cargo machinery spaces in 1.2.10, and the cargo motor room within the cargo area of any ship, shall be provided with a fixed fire-extinguishing system complying with the provisions of the FSS Code and taking into account the necessary concentrations/application rate required for extinguishing gas fires."

4.7.3.2 Chapter 5 of the FSS Code, i.e. Fixed gas fire -extinguishing systems, requires that the quantity of carbon dioxide for cargo spaces, unless otherwise provided, shall be sufficient to give a minimum volume of free gas equal to 30% of the gross volume of the largest cargo space to be protected in the ship, in paragraph 2.2.1.1.

4.7.3.3 On the other hand, NFPA 123) requires that the design quantity of carbon dioxide for

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4.7.3.3 On the other hand, NFPA 123) requires that the design quantity of carbon dioxide for hydrogen fire should be 75% or more of the gross volume of the protected space. The special requirement for an increased amount of carbon dioxide should be provided for carbon dioxide fire-extinguishing systems.

4.7.4 Features of hydrogen fire

Hydrogen burns at high temperature, but generally gives off less radiant heat than propane or other hydrocarbons (e.g. only about 10% of that radiated by an equal -sized propane flame). Although the heat radiated by a hydrogen flame is also relatively low compared to hydrocarbons, it is important to take into account the differences in heats of combustion,

* Electrical equipment used in hydrogen/air mixture should be, at least, the type of "II-C" and "T-1" as the group based on the maximum experimental safe gap for flameproof enclosures and the temperature class based

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based on the maximum experimental safe gap for flameproof enclosures and the temperature class based on maximum surface temperature, respectively, according to ISO/IEC 80079-20-14).

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burning rate and flame size. Hydrogen flames are colourless or nearly colourless. Both of these characteristics make it more difficult to detect a hydrogen fire. Even relatively small hydrogen fires are very difficult to extinguish. The only reliable approach to extinguish a fire is to shut off the source of hydrogen supply.

4.8 High pressure hazard

4.8.1 High pressure is a hazard common to hydrogen and other flammable gases listed in the Code. To prevent overpressure, the Code requires various measures such as pressure

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the Code. To prevent overpressure, the Code requires various measures such as pressure control and pressure design. Specifically, paragraph 8.2, in regard to the provision of pressure control of cargo tanks, requires fittings of pressure relief valves to the cargo tanks. Furthermore, paragraph 7.1.1 requires temperature control by the use of mechanical refrigeration and/or design to withstand possible increases of tempera ture and pressure. In addition, paragraph 15.2 specifies the filling limit of cargo tanks taking into account cargo volume increase by its thermal expansion. These requirements are applicable for hydrogen and no special requirement is considered necessary in this regard.

4.8.2 Boil-off may be a more significant issue for hydrogen than for LNG in particular when insulation properties have deteriorated. Means of handling boil -off gas should be carefully considered taking into account the following issues:

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insulation properties have deteriorated. Means of handling boil -off gas should be carefully considered taking into account the following issues:

.1 Re-liquefaction of hydrogen involves very specific and costly equipment. Cargo cooling in order to avoid boil-off shows the same kind of issues; and

.2 Notwithstanding the provision in paragraph 7.4.1 of the Code, thermal oxidation of hydrogen may be permitted in accordance with paragraph 1.3 of the Code.

4.8.3 The special requirements in these aspects are considered necessary.

4.9 Health hazard

4.9.1 Human safety concern under low temperature

With regard to the influences of cold hydrogen on persons' bodies, suitable protective equipment is effective. In this aspect, paragraph 14.1 of the Code requires suitable protective equipment taking into account the character of the products, therefore, no special requirement is considered necessary.

4.9.2 Static electricity

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equipment taking into account the character of the products, therefore, no special requirement is considered necessary.

4.9.2 Static electricity

Hydrogen ignition energy is very low and hydrogen can be easily ignitable by static electricity and due consideration should be given to this issue, in accordance with the requirement in the Code on suitable protective equipment.

4.9.3 Oxygen depletion and asphyxiation

Leakage of hydrogen may cause low level of oxygen and associated asphyxiation.

4.10 Wide range of flammable limits

4.10.1 Extinguishing hydrogen fire

4.10.1.1 As mentioned in paragraph 4.6, for flammable products the Code already requires elimination of sources of ignition, including use of electrical installations of appropriate types in order to minimize the risk of fire and explosion. No special requ irement is considered necessary with regard to ignitability of hydrogen.

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necessary with regard to ignitability of hydrogen.
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4.10.1.2 Furthermore, with regard to the wide range of flammable limits of hydrogen, the increased quantities of carbon dioxide as a fire -extinguishing medium should be specified as mentioned in paragraph 4.7. No additional special requirement is considered to be necessary with regard to the wide range of flammable limits of hydrogen.
4.10.2 Disposal of cold hydrogen gas
The wide flammability range makes disposal of cold hydrogen gas a major hazard. Cold plumes of released hydrogen may impede adequate dilution of hydrogen down to below 4% and may lead to flash-back to the vent from distant ignition sources outside safety -controlled areas. The low ignition energy and wide flammable range may present significant challenges.

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areas. The low ignition energy and wide flammable range may present significant challenges.

4.11 Prevention of dangerous purging operation

4.11.1 During cargo operations for maintenance, pipes and tanks should be purged with an inert gas or inert gases as illustrated in the figure below. For safety, due consideration should be given to temperature and boiling points of the inert gases. Residual pockets of hydrogen or the purge gas will remain in the enclosure if the purging rate, duration, or extent of mixing is too low. Therefore, reliable gas concentration measurements should be obtained at a number of different locations within the system for suitable purges. Temperature should also be measured at a number of locations. Oxidizing agents may exist in a hydrogen containing equipment, specifically: air, cold box atmospheres containing air diluted with nitrogen, or oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.

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oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.

4.11.2 There are special measures that may need to be put in place in order to mitigate the hazards, e.g. air should be eliminated by nitrogen purge prior to introduction of hydrogen into cargo piping or processing equipment. Nitrogen should then be eliminated by hydrogen purge, where there is a possibility of its solidification in the subsequent process.

Figure 1

References in part A

1 ISO/TR 15916:2015, Basic consideration for the safety of hydrogen systems (ISO)

2 American Institute of Aeronautics and Astronautics, "Safety Standard for Hydrogen and Hydrogen Systems (Guide to Safety of Hydrogen and Hydrogen Systems)", 2005 (AIAA)

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